Catherine Lin

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EDUCATION

Bridgeland High School (CEEB: 440281) Cypress, Texas

Cumulative GPA: 6.8298/7 | ***Non-Weighted GPA:*** 3.9556/4 | ***Rank:*** 29/887

ACT: 36 Composite (English: 35, Math: 36, Reading: 36)

RESEARCH

Research under Professor Suleyman Uludag, University of Michigan Nov 2025 – Present

"*From Execution to Governance: How AI Reshapes Human Roles Across Career Stages"*

- Synthesized 41 academic, policy, and industry sources in a structured narrative literature review examining AI's effects on human skills, professional responsibilities, education, and governance.
- Developed a conceptual framework describing shifts toward evaluation at entry level, integration at mid-career, and governance at senior level, distinguishing empirical findings from proposed role changes.
- Independently researched, wrote, and revised a structured narrative literature review under the guidance of Professor Süleyman Uludağ that is submitted to a journal for review.

Pioneer Academics Jun 2025 – Aug 2025

AI and Workforce Readiness for Computer Scientists Across Career Stages

- Examined shifts in computer science careers from the pre-AI era through future workforce projections, focusing on how emerging technologies are changing pathways into and through the field.
- Synthesized findings from academic literature, industry reports, and workforce datasets to evaluate changes in skill requirements, job market accessibility, AI adoption, and the need for continuous upskilling.

Science Internship Program (SIP) – University of California, Santa Cruz Jun 2024 – Aug 2025

CPM-13: AI assistants for digital boundary communication and management

- Conducted Human-Computer Interaction (HCI) research on AI-powered digital boundary management systems, reviewing literature and analyzing strategies to improve user well-being through notification filtering and prioritization.
- Collaborated with a research team to design and prototype AI assistant concepts that help users establish, manage, and communicate digital boundaries, promoting healthier technology use and work-life balance.
- Invited to continue the project beyond the internship as a co-author on a research paper, "Reconfiguring AI Mediation as Inquiry Through Youth Participatory Action Research," currently submitted to the CHI 2027 Conference.

INDEPENDENT PROJECT

Cofounder & CEO**, Sound of the Future** May 2025 – Present

- Founded a hearing-health and Deaf/hard-of-hearing advocacy nonprofit; expanded to 5 U.S. and 2 international chapters, reached 240K+ people through community initiatives and media outreach, and distributed 1,500+ care packages.
- Led school and community programs, hearing-health outreach at STEMGO PRO TECHFEST and Girls Gotta Shop Market, and educational podcasts across Spotify, iHeartRadio, and YouTube.
- Partnered with the Hearing Health Institute on care package supplies and a monthly blog for its news and website.
- Raised $3,000+ through Benevity and market sales; secured sponsorships from Costco, Home Depot, Hopdoddy, Half Price Books, and JK Hair Salon; invited to a KPRC 2 livestream interview and featured on KPFT Houston radio.

Independent Research**, AI & Employment Research Initiative** May 2026 – Present

- Designed and conducted structured interviews with HR professionals and industry employees on how AI is reshaping hiring, job displacement, and emerging skill demands.
- Coded and analyzed qualitative interview data to identify patterns in employment expectations and workforce preparation; presented findings on a self-coded public website.

Independent Research, **AI & Asian Restaurant Technology Study** June 2026 – Present

- Interview Asian restaurant owners and managers to identify gaps in existing AI systems, including authentic dish names, regional accents, multilingual communication & technology adoption; Translate user research into an AI-powered restaurant management app designed around owners' linguistic and cultural needs rather than requiring them to adapt to existing technology.
- Prototype multilingual menu management, cross-platform ordering, automated translation, cost/profit analysis, culturally aware recommendations & AI-assisted marketing tools.
- Bilingual interview feature at the Southern News Group (Biggest Asian American Multi-Media Group in the US.)

EXTRACURRICULAR ACTIVITIES

Team Captain**, Vex Robotics** May 2024 – Present

- Captain a nationally competitive robotics team, overseeing engineering design, build strategy, project planning, task delegation & competition preparation.
- Direct on-field strategy by analyzing scouting data, robot capabilities & alliance compositions to develop match plans and make real-time decisions under pressure.

- Oversee engineering documentation and mentor elementary/middle school VEX teams in design, programming, engineering notebooks & competition strategy.
- Led the team to 3 consecutive VEX World Championship qualifications, 25+ Tournament Championships, 10+ Excellence Awards, 15+ Robot Skills Championships, 2 Texas Amaze Awards & 2 Signature Event Finalist finishes.

***Social Director,* Science Olympiad** Aug 2022 – Present

- Manage communications and community-building for an 85+ member team, including social media, team events, competition logistics, travel itineraries & the after-school café.
- Selected among 15 of 85+ members to represent Bridgeland at the Texas State Tournament for four consecutive years.
- Competed in 30+ events and earned 20+ medals, including 2nd in Tower at Texas State and 7th in Bungee Drop at the Golden Gate Invitational.

***Reporter & Leadership Team Officer,* Technology Student Association (TSA)** Aug 2024 – Present

- Support chapter operations through executive communications, meeting presentations, competition logistics, member activities & mentorship of younger competitors.
- Competed across engineering, data science, photography, presentation & design; earned 10+ regional and 8+ Texas State awards, including 1st place in four Unique to Texas engineering events.

***Builders Club Advisor,* Key Club** Aug 2024 – Present

- Lead biweekly leadership and community-service programming for 100+ middle school students across two partner schools.
- Organize service projects and mentor students in leadership, teamwork & civic engagement while supporting volunteer coordination for 150+ Key Club members.

***Calculator Applications Officer,* Math UIL** Aug 2025 – May 2026

- Mentored students through weekly Calculator Applications practices, competition strategy & preparation for invitational, regional and state contests.
- Built a centralized resource system for practice tests, study guides, schedules & competition materials; coordinated competition logistics and communications.

Rice Elite Tech Camp Summer 2025

- Applied engineering, programming, AI & robotics concepts through collaborative, project-based STEM design challenges.

Code 2 College Jun 2026 – Present

- Completed a 9-week web/programming course covering foundational and advanced Python alongside professional-development workshops in time management, career skills & portfolio development.
- Built Python applications including Hangman, an Escape Room game & an inventory-management system using functions, loops, dictionaries, file handling and object-oriented programming; debugged and iteratively refined projects.

***Leadership Team Member,* Pioneer Academics Alumni & Social Media Committee** Aug 2025 – Present

- Create educational content for Pioneer's official social media on research opportunities, applications, productivity & lessons from my own research experience.
- Mentor current researchers through alumni workshops, advising on research design, academic writing, faculty communication & the independent research process..

Freelance Photographer Aug 2021 – Present

- Photograph portraits, cityscapes, nature & special occasions; maintain a VSCO portfolio of original work and accept paid client bookings.
- Operate an eBay storefront selling original photography prints, managing online sales, customer service, order fulfillment & storefront operations.

***Founder*, DoubleS (Consumer Hardware Startup)** Jun 2026 – Present

- Founded a startup developing an ultra-thin, rechargeable tracking sticker for everyday items; led early prototyping integrating Bluetooth Low Energy (BLE), wireless charging, and a piezo buzzer.
- Conducted market validation, generating 14,000+ responses and expressions of consumer interest to assess demand and guide product development.
- Selected for Startup Live Launch at Startup Revolution AI Summit 2026, a program attracting 200 startup applications, with an invitation to present DoubleS on the main stage in Skopje.

WORK EXPERIENCE

***E-Commercial Resale Business,* YUBAI** Summer 2023, 2024, 2025, 2026

- Provided bilingual English/Mandarin customer support, resolving inquiries and order issues throughout the purchasing process.
- Supported e-commerce operations through product photography, customer outreach, order organization, QR-code systems, troubleshooting & fulfillment.

***Restaurant Operations Assistant,* Fang's Bistro** June 2026 – Present

- Created and manage the restaurant's Instagram; produce promotional content and redesigned menus to strengthen its visual identity and customer outreach.
- Support front-of-house operations by greeting customers, answering phones, processing orders & coordinating DoorDash and third-party delivery services.

COMMUNITY SERVICE

***Founder & Photographer,* Freedom Frame** April 2026 – Present

- Founded a global photography/storytelling initiative documenting women entrepreneurs, leaders, artists & changemakers; recruit photographers, writers & ambassadors to expand coverage across communities.
- Photograph and interview featured women; direct visual storytelling, social media, partnerships & publication of their stories across digital platforms.

***Houston Chapter Head,* Between the Lines** April 2026 – Present

- Lead the Houston chapter of a student-led initiative highlighting Asian and Asian American athletes; organize community pickleball sessions and a local tournament, coordinating participants, logistics & 3D-printed prizes: over 200+ participants
- Photograph and interview local athletes and contribute written and visual features to the organization's annual magazine, documenting their achievements, identities & experiences in sports.

HONORS & AWARDS

VEX Robotics

- **Signature Event Tournament Finalist**, **Ranking #2/75 (National),** Space City 2025
- **Signature Event Tournament Finalist, #1/90 teams (National),** Robot Rodeo 2026
- **Signature Event Think Award (National),** Space City 2025
- **Texas Region 3 Tournament Champion,** qualifying for the **VEX Robotics World Championship (State)** 2024
- **Texas Region 3 Robot Skills Champion (State)** 2024
- **2× Texas Region 3 Amaze Award**, qualifying for the **VEX Robotics World Championship (State)** 2025, 2026
- **Tournament Champion (National),** CREATE U.S. Open 2024
- **Robot Skills Champion (National),** CREATE U.S. Open 2024
- **25+ Local Tournament Championships (District)** 2024, 2025, 2026
- **10+ Local Excellence Award (District)** 2024, 2025, 2026
- **15+ Local Skills Championships (District)** 2024, 2025, 2026

Technology Student Association (TSA)

- **3× 1st Place Mechanical Engineering (State)** 2024
  - Working Drawings of a Single or Multiple Parts with a Pictorial View
  - Pictorial View Only of 6-12 Mechanical Parts
  - Pictorial View Only of 2-5 Mechanical Parts
- **5th Place Photography - People - Portrait (State)** 2026
- **1st Place Finish Model of a Residential Structure not listed (State)** 2026
- **TSA Lonestar Degree & Texan Degree** 2025
- **1st place Mechanical Engineering (Regional)** 2024
  - Working Drawings of a Single or Multiple Parts with a Pictorial View
  - Pictorial View Only of 6-12 Mechanical Parts
  - Pictorial View Only of 2-5 Mechanical Parts
- **1st Place Data Science and Analytics(Regional)** 2025
- **3rd Place Senior Solar Sprint (Regional)** 2025
- **1st Place Finish Model of a Residential Structure not listed (Regional)** 2026
- **1st Place Prepared Presentation (Regional)** 2026
- **1st Place Photography (Regional)** - People - Portrait 2026
- **3rd Place Photography (Regional)** - People - Children (Regional) 2026

Science Olympiad

- 2nd **Place, Tower** *(Texas State Tournament; 2nd of 50 teams)* 2025
- 7th **Place, Bungee Drop** *(National Invitational; 7th of 70 teams)* 2026
- 5x **Regional Placements** (Different Events) *(Regional)* 2024, 2025, 2026
- 8x **Regional Placements** (Different Events) *(District)* 2024, 2025, 2026
- 15x **Local Placements** (Different Events) *(District)* 2024, 2025, 2026

USACO Bronze Division *(National)* 2025

College Board AP Scholar with Distinction 2025, 2026

Invitation to Become a Columnist at Southern News Group (Biggest Asian American Multi-Media Group in the US) 2026

SKILLS

Languages: English (Native), Mandarin Chinese (Native)
Programming: Python, Java, C++, OpenAI API Integration
Engineering: Autodesk Inventor, Fusion 360, AutoCAD, Prompt Engineering